this optimization that can result in linearly encoded representations. The goal here is to highlight the underlying dynamics that prefer linear representations. It is worth noting that we choose to study the role of gradient descent instead of the more computational tractable stochastic gradient descent used in experiments for analytic simplicity. However, according to classical stochastic approximation theory [KY03], with a sufficiently small learning rate, the additional stochasticity is negligible, and stochastic gradient descent will behave highly similarly to gradient descent [Wu+20].

The key observation is that there is a hidden binary classification task. Let's consider the simple three-variable example. Suppose instead of predicting all possible conditional distributions of latents, the model only learns $p(\cdot|c_1 = 1)$ and $p(\cdot|c_1 = 0)$. Denoting $v = g(1,1,1) - g(0,1,1), w_0 = f(0,\diamond,\diamond), w_1 = f(1,\diamond,\diamond)$,

$$\frac{\hat{p}(0,1,1|c_1 = 1)}{\hat{p}(1,1,1|c_1 = 1)} = \exp(-w_1^T v) \approx 0,$$
$$\frac{\hat{p}(1,1,1|c_1 = 0)}{\hat{p}(0,1,1|c_1 = 0)} = \exp(w_0^T v) \approx 0$$

Equivalent, the model is trying to optimize the following loss:

$$L(w_1, w_0, v) = \sum_{i=0}^{1} \ell(-y_i w_i^T v)$$

where $y_0 = 1, y_1 = -1$ and $\ell(x) = \exp(-x)$. Intuitively, if w_0 and w_1 are fixed, then this is a binary classification task with exponential loss. It is known that gradient descent under this setting would converge to the max-margin solution [Sou+18], which makes the direction of v unique.

The following theorem makes this intuition precise. It states that by optimizing a specific subproblem of the next token predictions using gradient descent with *embeddings fixed*, latent unembedding representations will be encoded linearly. For vectors u, v, denote $\cos(u,v) = \frac{\langle u,v\rangle}{||u||\cdot||v||}$.

Theorem 4 (Gradient descent with fixed embeddings)**.** *Fix $i \in [m]$. Let $\widehat{\mathscr{D}} = \{d^{\diamond}_{(i\to 1)}, d^{\diamond}_{(i\to 0)}\}$ where $d^{\diamond} = [\diamond, \ldots, \diamond]$ and $\Delta_{c,i} = g(c_{(i\to 1)}) - g(c_{(i\to 0)})$. Suppose the loss function is the following:*

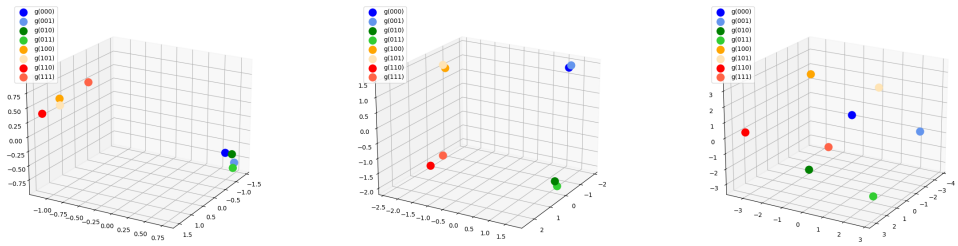
$$L(\{\Delta_{c,i}\}_c, f(d^{\diamond}_{(i\to 1)}), f(d^{\diamond}_{(i\to 0)})) = \sum_{c\in\overline{\mathscr{C}}} \left(\exp(-\Delta_{c,i}^T f(d^{\diamond}_{(i\to 1)})) + \exp(\Delta_{c,i}^T f(d^{\diamond}_{(i\to 0)}))\right)$$

where for all $c \in \overline{\mathscr{C}}$, $c_i = 1$ and $f(d^{\diamond}_{(i\to 1)})) \neq f(d^{\diamond}_{(i\to 0)})$. Then fixing f and training g using gradient descent with the appropriate step size, we have

$$\lim_{t\to\infty} \cos(\Delta^t_{c^1,i}, \Delta^t_{c^2,i}) = 1$$

for any $c^1, c^2 \in \overline{\mathscr{C}}$ where the superscript t is meant to represent vectors after t number of iterations.

All proofs are deferred to Appendix C. The theorem says that under gradient descent with the exponential loss function and embeddings fixed, the unembedding vector differences for a fixed concept are all aligned in the limit. It is worth mentioning that although the loss function is a bit simplified, it does represent the significant subproblem of the optimization in terms of linear geometry. The reasons are two-fold: (1) Because conditional probabilities are learned with the softmax function, there is an extra degree of freedom. In other words, one does not need to use all the logits to estimate distributions, just the differences between them. (2) Due to the nature of the exponential function, the closer the ratios between conditional distributions are to zero, the larger the distances between unembeddings need



(a) $\widehat{\mathscr{D}} = \{(0,\diamond,\diamond),(1,\diamond,\diamond)\}$ **(b)** $\widehat{\mathscr{D}} = \{(0,0,\diamond),(0,1,\diamond),(1,0,\diamond),(1,1,\diamond)\}$ **(c)** $\widehat{\mathscr{D}} = \mathscr{D}$

Figure 2: Unembedding representations form different clusters after training on various conditioning sets. This figure shows unembedding representations after learning different sets of conditional distributions.

to be. Therefore, learning to predict these zero conditional probabilities would dominate the direction of concept representations.

The above theorem assumes that the embedding space is fixed. It turns out that gradient descent will also have the tendency to align embedding and unembedding space when they're not fixed and are trained jointly, as we will show next. First, one can observe that if the 0 conditional probabilities are learned to a reasonable approximation, then the inner product of embedding and unembedding steering vectors of the same concept should be large (Proposition 11) or in other words, the exponential of the inner product is approaching infinity. But since there are different ways that this inner product can reach infinity, a natural question to ask is what happens when one purely optimizes this exponential learning objective. Theorem 12 shows that gradient descent, when trained to minimize the exponential of negative inner product of two vectors, would align these two vectors.

Finally, using Theorem 4 and Theorem 12, we obtain the following statement, which is one of our core contributions.

Theorem 5 (Gradient descent aligns representations)**.** *Under the conditions of Theorem 4, if one further assumes that* $\{\Delta_{c,i}\}_c, f(d^{\diamond}_{(i\to 1)})), f(d^{\diamond}_{(i\to 0)})$ *are initialized to be mutually orthogonal with the same norm, then optimizing* f *and* g *using gradient descent, we have*

$$\lim_{t\to\infty} \cos(\Delta^t_{c^1,i}, \Delta^t_{c^2,i}) = 1$$
$$\lim_{t\to\infty} \cos(\Delta^t_{c^1,i}, f^t(d^{\diamond}_{(i\to 1)}) - f^t(d^{\diamond}_{(i\to 0)})) = 1$$

for any $c^1, c^2 \in \overline{\mathscr{C}}$ *where the superscript* t *is meant to represent vectors after* t *number of iterations.*

While stated generally as a result of the properties of gradient descent, Theorem 5 says that when the latent conditional models are trained on a subproblem of next token prediction with reasonable initializations (see below remark), implicit bias of gradient descent will align the embedding and unembedding vectors as well as aligning among the unembedding vectors. This can manifest as linear representations in the final trained LLMs such as LLaMA-2 [Tou+23]. Although the theorem addresses only a specific subproblem, it notably suggests, as discussed earlier, the underlying bias and dynamics that contribute to the emergence of linear representations. That is, if one solely optimizes this subproblem, one can get perfect linear representations. We verify this result empirically in Section 5.

Remark 6. The initial condition assumption in the theorem statement seems unnecessary as evidenced by our experiments in Section 5. On the other hand, if one initializes vectors

with high dimensional multivariate Gaussians, then the assumption will be approximately satisfied with high probability [Das99].

The role of learning various conditional distributions So far, we have only discussed learning conditional distributions for a single concept. In general, due to the presence of zero probabilities in various conditioning contexts, learning these conditional probabilities is, in some sense, learning different forces to push sets of representations far apart because one needs relatively large inner products to get exponents to be close to zero.

Figure 2 gives an illustrative example with three concept variables that shows how learning different subsets of conditional probabilities would push various subsets of unembeddings in different ways. For example, if the model exclusively learns conditional probabilities for C_1, the unembedding representations would form clusters based on the values of C_1. Similarly, if the model learns conditional probabilities for both C_1 and C_2, the unembedding representations would cluster based on the values of both C_1 and C_2. However, in the second case, the concept directions of C_1 and C_2 are coupled. As shown in Figure 2(b), $g(110) - g(100)$ is parallel with $g(111) - g(101)$ but not with $g(010) - g(000)$. To decouple them, one needs a bigger $\widehat{\mathscr{D}}$ as in Figure 2(c). That said, it seems that a complete set of $\mathscr{D}$ is not always necessary (Section 5).

An analogy would be to view each binary vector as a combination of "electrons" where the values represent positive or negative charges. Learning probabilities under different conditioning contexts can be likened to applying various forces to manipulate the positions of these electrons. This is similar to the Thomson Problem in chemistry and occurs in other contexts of representation learning as well [Elh+22].

4 Orthogonality

A surprising phenomenon observed in LLMs like LLaMA-2 [Tou+23] is that the Euclidean geometry can somewhat capture semantic structures, despite the fact that the Euclidean inner product is not identified by the training objective [PCV23]. Consequently, one notable outcome is the tendency for semantically unrelated concepts to be represented as almost orthogonal vectors in the unembedding space. In this section, we study this phenomenon under our latent conditional model. Based on our underlying Markov random field distribution, we define unrelated concepts to be those separated in G_C.

Theorem 7. *Let $\widehat{\mathscr{C}} = \mathscr{C}$ and $\widehat{\mathscr{D}} = \mathscr{D}$. Assuming $p(c) > 0$ for any $c \in \mathscr{C}$ and C_i and C_j are two latent variables separated in G_C. Given any binary vector $c \in \mathscr{C}$, there exists a subset $\mathscr{D}_c \subset \mathscr{D}$ such that $d_i = \diamond$ and $p(c|d) > 0$ for any $d \in \mathscr{D}_c$. If one further assume that $\hat{p}(c|d) = p(c|d)$ for any $d \in \mathscr{D}_c$, then*

$$g(c_{(i\to 1)}) - g(c_{(i\to 0)}) \perp f(d_{(j\to c_j)}) - f(d_{(j\to\diamond)})$$

for any $d \in \mathscr{D}_c$.

All proofs are deferred to Appendix D. As suggested in Section 3.2 and verified empirically in Section 5, under the latent conditional model, representations are not only linearly encoded but also frequently aligned between embedding and unembedding spaces, which implies the following corollary on the orthogonal representation of unrelated concepts.

Corollary 8. *Under the conditions of Theorem 7, if concept C_i and C_j have matched-representations, then*

$$g(c_{(i\to 1)}) - g(c_{(i\to 0)}) \perp g(c_{(j\to 1)}) - g(c_{(j\to 0)})$$

for any $c \in \mathscr{C}$.